

Development of a smartphone SDK for obfuscating location data

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Abstract—In today’s data-driven economy, location data is of immense value. It is used to gain meaningful crowd insight and optimize various sectors of society, ranging from tourism to retail. However, the methods used to collect these valuable location traces often occur without the user’s explicit informed consent. The high precision of location data and the frequency of reports pose significant privacy risks to individuals. These risks include not only the inference of sensitive locations, such as home, work, and leisure activities, but also the construction of social graphs. In the hands of malicious actors, this information could lead to blackmail, coercion, or hate crimes. To address these privacy concerns, this thesis proposes a privacy-friendly alternative to collect valuable location traces. A novel Android library is developed that obfuscates both locations and timestamps by introducing noise, ensuring that individual traces can no longer be linked to a specific user. This prevents sensitive information from leaking while preserving the utility of the data. Importantly, the library operates entirely on-device, ensuring that the real, accurate traces never leave the device. To achieve this, this thesis explores the Android ecosystem on lifecycles and background tasks to design and implement advanced location obfuscation algorithms within the library. A set of experiments are conducted to demonstrate the effectiveness of the solution. Among other evaluations, this thesis compares the proposed library to the Approximate Location feature introduced in Android 12, offering information on advances in mobile privacy measures.

Index Terms—Android library, location obfuscation, location tracking

I. INTRODUCTION

A. Context and problem definition

The central problem is that the real location of a user is usually sent directly to the server and stored there without any form of obfuscation. As a result, location data remains available in its original and unobfuscated form, entails significant privacy risks. Furthermore, this data is often sold on to data brokers without further processing.

In order to guarantee the privacy of users, several conditions must be met. First, data transmission to the server must be secured against interception. Second, the server itself must be adequately protected against data leaks. In addition, it is essential that stored location data is stored in an obfuscated manner and that any resale of this data also takes place only in obfuscated form. Finally, strict control over the parties that gain access to this data is necessary to prevent misuse.

The presence of these potential vulnerabilities makes it difficult for users to have confidence in the current processing of location data.

In addition, the steep learning curve of relevant Android technologies poses an additional challenge. This can lead to errors or security gaps in the implementation, especially with limited experience or tight deadlines. A well-designed library that reduces complexity for developers and provides robust security measures as standard can significantly reduce these risks.

II. RELATED WORK

A. Location Privacy Risks

The collection of high-resolution and high-frequency location data introduces significant privacy risks. Research by Kevin De Boeck has demonstrated that seemingly anonymized datasets can be used to infer private places such as homes, workplaces, and vacation patterns. Even sparse traces can reveal sensitive information, especially in rural areas where few streets exist within a given radius. Malicious actors may use this data for profiling, blackmail, or even targeted crimes. [2]

These risks are not purely theoretical. In practice, data brokers like Gravy Analytics have experienced massive data breaches that exposed millions of users’ location information [1]. Additionally, location-based dating apps have historically leaked users’ exact positions by allowing triangulation through proximity features [4]. As a result, government and military bodies have flagged such applications as national security concerns.

B. Existing Solutions

To mitigate these risks, several obfuscation strategies have been proposed and implemented. Android 12 introduced the concept of “Approximate Location,” allowing users to restrict apps to a coarser location estimate. However, the mechanism used is closed-source, non-configurable by developers, and offers limited transparency.

Academic work has proposed more robust, configurable solutions. One such approach is the concept of “privacy circles,” used in prior work by Toon Dehaene. His system introduced random displacement within a radius and ensured

that location points could not be averaged to recover the original coordinates. His implementation in Rust allowed for batch obfuscation on large datasets but required off-device processing. [3]

This thesis builds upon such prior work by implementing the obfuscation pipeline directly on Android devices, ensuring that accurate location data never leaves the device and that obfuscation is integrated seamlessly into the data collection process.

III. SYSTEM DESIGN

A. Requirements

The design of the Android SDK is guided by both technical and privacy requirements. Functionally, the SDK must:

- Collect location and timestamp data at configurable intervals.
- Apply a robust obfuscation method before any transmission or storage.
- Respect Android's lifecycle, background limitations, and permission model.

From a developer usability perspective, the SDK should:

- Offer a simple integration flow with minimal lines of code.
- Allow developers to plug in their own obfuscation logic.
- Provide debug tools for validation of both obfuscated and raw traces.
- Automatically handle permission checks and WorkManager setup.

B. Architecture

The system consists of three main components:

- LocationManager: A singleton [5] responsible for configuration, lifecycle management, and coordination between services.
- LocationObfuscator: A WorkManager Worker implementation that handles background location polling, either periodically or via foreground services.
- LocationWorker: An abstract interface that allows the injection of custom obfuscation algorithms. A default implementation is provided based on privacy circles.

The SDK separates concerns clearly: location retrieval is decoupled from obfuscation logic, and both are decoupled from data transmission. This ensures flexibility, maintainability, and extensibility for future use cases.

IV. IMPLEMENTATION

A. Obfuscation Algorithm

The default obfuscation algorithm adapts Toon Dehaene's privacy circle approach for real-time data streaming. The core logic is:

- If a new location lies within an existing privacy circle, reuse the circle's centre.
- If outside all circles, generate a new circle that's displaced by a random angle and random distance between 0 and the currently used radius.

- Respect a minimum time interval between emissions to avoid over-reporting.
- Prevent jitter by favouring reuse of the most recently active circle over the closest one. Given the point lies within the most recently active circle.

This algorithm minimizes the risk of location recovery via averaging or triangulation. All operations occur locally, and no raw data is retained or transmitted unless in debug mode. Storage of circles is done via Android's `filesDir`, which is sandboxed per application. So only if the system is compromised a data leak could occur.

B. Android-specific challenges

Due to Android's strict limitations on background execution, the SDK uses WorkManager as its backbone. It supports both foreground services (with a visible notification) and periodic jobs (every 15 minutes max, running up to 10 minutes).

Permission handling is implemented using `ActivityResultContracts.RequestPermission`. The library checks and requests permissions for both coarse and fine location access and guides users to system settings when needed.

To prevent memory leaks and lifecycle crashes, the LocationManager was made a singleton so the user couldn't have two instances that together could crash the system. All blocking calls are offloaded to thread or the LocationWorker. As Android enforces no blocking operations on the main thread.

V. EVALUATION

A. Experimental Setup

To evaluate the SDK, a controlled test route was followed through Ghent. The route included:

- Departing from KU Leuven Technology Campus,
- Stopping at the Korenmarkt (long idle period),
- Passing De Krook (short idle period),
- Boarding a train at Dampoort towards Lokeren.

This route enabled the evaluation of the SDK under different speeds, environments (urban, indoor), and stop durations. Four Android phones were used to record traces, each with a different configuration of variables. For the obfuscation evaluation, 42 configurations were tested (7 radii $\times$ 6 intervals). All traces were visualized using Jupyter notebooks and the Folium mapping library. Obfuscation results were compared against raw traces to evaluate privacy, continuity, and utility.

B. Key Findings

- Privacy vs Utility Trade-off: Larger privacy radii (e.g., 1600m) resulted in fewer, broader circles, which significantly improved anonymity but reduced trace granularity. Conversely, smaller radii retained more precise path information at the cost of lower privacy.
- Hardware Influence: Indoor locations (e.g., train or buildings) showed more location jitter, but the algorithm's reuse of prior circles prevented excessive data fluctuation.
- Approximate vs Fine Location: Android's built-in coarse location showed signs of both stochastic and deterministic

obfuscation, but the proposed SDK offered more consistency and developer control.

These results confirm that on-device obfuscation is feasible and effective without requiring backend involvement, and can be tuned according to the intended level of privacy and analytical utility.

VI. CONCLUSION

This thesis demonstrates that it is possible to obfuscate location data on-device without significantly affecting its usability for analysis. By developing an Android library that distorts locations and timestamps via noise and privacy circles, it is prevented that individual users can be identified via their movement patterns. The core of the solution lies in preserving data utility: aggregated insights, such as mobility trends or crowd analyses, remain possible, while sensitive individual information is protected.

A key success is the complete independence from external servers. All processing takes place on the device itself, so that the original, accurate location never leaves the device. This eliminates the risk of data leakage during transmission or storage. Technically, this was achieved by a modular design, where the library seamlessly integrates with Android's Fused Location Provider and WorkManager for reliable background processing. In addition, the implementation of a standard obfuscation algorithm based on privacy circles provides a balance between anonymization and accuracy, inspired by previous research.

The library not only lowers the implementation barrier for developers by abstracting complex privacy techniques, but also contributes to a paradigm shift in data management. Instead of treating privacy as an afterthought, protection becomes an inherent part of the design. This aligns with growing regulatory requirements and societal demand for transparent data practices.

While further optimization remains possible, this approach shows that technical solutions can play a key role in reconciling data innovation and individual rights. By making privacy-by-design accessible, this thesis takes a step towards a future in which technology does not come at the expense of user control, but rather enhances it.

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